Chapter 4: Model checking in the Cox regression model — Solutions

STA310 Survival Analysis (Master's in Applied Biostatistics)

Table of contents

1 Residuals for the Cox regression model (Book §4.1) 1
1.1 Cox–Snell residuals 1
1.2 Martingale residuals 2
1.3 Deviance residuals 2
1.4 Schoenfeld residuals 3
2 Assessing model fit (Book §4.2) 3
2.1 The cumulative hazard plot of the residuals 3
2.2 Checking the functional form of covariates 4
2.3 Identifying poorly fitted observations 4
3 Influential observations (Book §4.3) 5
3.1 Delta-betas and influence 5
4 Checking proportional hazards (Book §4.4) 5
4.1 The log-cumulative hazard plot 5
4.2 Schoenfeld residuals and the zph test 6
4.3 What to do when proportional hazards fails 6
5 Putting it together 7
5.1 A misleading diagnostic conclusion 7
6 Mini-lab: checking a Cox model in R 7
6.1 Detecting non-proportional hazards 7

Chapter 4 covers the diagnostics used to assess whether a fitted Cox model is adequate: the various residuals (Cox–Snell, martingale, deviance, Schoenfeld, score), their use in checking overall fit and the functional form of covariates, the detection of influential observations, and — most importantly — checking the proportional hazards assumption.

1 Residuals for the Cox regression model (Book §4.1)

1.1 Cox–Snell residuals

Type: Concept & notation

The Cox–Snell residual for the ith individual is

$$r_{Ci} = \exp(\hat{\beta}' x_i)\, \hat{H}_0(t_i).$$

(a) Express r_{Ci} in terms of the estimated cumulative hazard (or survivor function) of the ith individual.
(b) If the fitted model is correct, what distribution do the Cox–Snell residuals approximately follow? State its mean, and say whether it is symmetric.
(c) What happens to the residual of a right-censored individual, and how does this complicate the use of the residuals for checking?

Solution.

(a) $r_{Ci} = \hat{H}_i(t_i) = -\log \hat{S}_i(t_i)$: the estimated cumulative hazard (or minus the log of the estimated survivor function) for the individual, evaluated at their observed survival time.
(b) They approximately follow a unit exponential distribution, with mean 1 (and variance 1). This distribution is highly skew and non-negative — the residuals are not symmetric about zero, unlike residuals in linear regression.
(c) If t_i is censored, the residual is itself censored: we only know that the true residual exceeds the value computed at the censoring time. The residuals therefore form a censored sample from the unit exponential distribution.

1.2 Martingale residuals

Type: Concept & notation

Builds on: Chapter 3

The martingale residual is $r_{Mi} = \delta_i - r_{Ci}$, where δ_i is the event indicator.

(a) Interpret r_{Mi} as the difference between two quantities.
(b) What is the range of possible values, and what sign do censored observations always have?
(c) What property, shared with residuals from linear regression, do the martingale residuals have?

Solution.

(a) It is the difference between the observed number of events for individual i over $(0, t_i)$ — namely δ_i, which is 0 or 1 — and the estimated expected number of events over that interval, $r_{Ci} = \hat{H}_i(t_i)$.
(b) The range is $(-\infty, 1]$. For a censored observation $\delta_i = 0$, so $r_{Mi} = -r_{Ci} < 0$: censored residuals are always negative.
(c) They sum to zero (and, in large samples, are approximately uncorrelated with expected value zero) — like residuals in linear regression.

1.3 Deviance residuals

Type: Concept & notation

(a) Why were deviance residuals introduced, given that martingale residuals already existed?
(b) In broad terms, how do they relate to martingale residuals?
(c) Do deviance residuals sum to zero?

Solution.

(a) Martingale residuals are skew (asymmetric) even when the model is correct, which makes plots hard to interpret. Deviance residuals are transformed to be roughly symmetric about zero when the model is appropriate.
(b) They are a sign-preserving transformation of the martingale residuals: large negative martingale values are shrunk towards zero, and values near the upper limit of 1 are expanded, producing a more symmetric distribution.
(c) No — unlike martingale residuals, deviance residuals need not sum to zero.

1.4 Schoenfeld residuals

Type: Concept & notation

Builds on: Chapter 3

The Schoenfeld residual for individual i and explanatory variable X_j is $r_{Sji} = \delta_i \left(x_{ji} - \hat{a}_{ji}\right)$, where $\hat{a}_{ji}$ is a risk-set-weighted average of X_j among the individuals at risk at time t_i.

(a) How many Schoenfeld residuals are there for each individual, compared with the Cox–Snell or martingale residuals?
(b) Why are the residuals zero for censored observations, and what does $\hat{a}_{ji}$ represent?
(c) For what specific purpose are the (scaled) Schoenfeld residuals especially useful?

Solution.

(a) One per explanatory variable, not one per individual: r_{Sji} is the residual for individual i with respect to covariate X_j, so a model with p covariates gives p residuals for each person.
(b) They contain the factor δ_i, so censored observations ($\delta_i = 0$) give zero. $\hat{a}_{ji}$ is the "expected" value of X_j for someone dying at t_i under the model — the risk-set average of X_j. The residual compares the observed x_{ji} with this expectation.
(c) Checking the proportional hazards assumption: plotting scaled Schoenfeld residuals against (a function of) time, and the associated Grambsch–Therneau (zph) test.

2 Assessing model fit (Book §4.2)

2.1 The cumulative hazard plot of the residuals

Type: Interpret figure

A common diagnostic is a plot of $\hat{H}(r_{Ci}) = -\log \hat{S}(r_{Ci})$ — the estimated cumulative hazard of the Cox–Snell residuals — against the residual values r_{Ci}.

(a) What is being plotted, and why does a straight line through the origin with unit slope indicate a good fit?
(b) You obtain a plot that curves systematically away from that line. What do you conclude?
(c) Why is this plot regarded as insensitive — not to be relied on by itself?

Solution.

(a) It plots the estimated cumulative hazard of the Cox–Snell residuals against the residual values. If the residuals follow a unit exponential distribution, their cumulative hazard is $H(r) = r$,

a straight line through the origin with unit slope; a straight line therefore supports model adequacy.

(b) A systematic departure from the line suggests the model is misspecified and needs to be modified.

(c) The residuals only approximately follow a unit exponential distribution (the approximation depends on the estimated βs and baseline hazard), and the plot can look linear even when the model is seriously wrong; the null model always gives a straight line. It must be supplemented by other diagnostics.

2.2 Checking the functional form of covariates

Type: Diagnostics & model critique

(a) Describe how martingale residuals from the null model can be used to check the functional form of a continuous covariate.

(b) What shape in the plot indicates that a linear term is adequate, and what would indicate that a transformation is needed?

(c) In the myeloma example, the smoothed martingale plot against Bun was roughly straight, but the plot against $\log(\text{Bun})$ was curved. What does this suggest about how Bun should enter the model?

Solution.

(a) Fit the null model (no covariates), obtain its martingale residuals, and plot them against the covariate, with a smoothed curve superimposed. The plot reveals the functional form the covariate should take in the model.

(b) A straight line indicates a linear term suffices. A curved pattern indicates non-linearity, suggesting a transformation (e.g. a logarithm) or a fractional polynomial.

(c) A straight line for Bun but curvature for $\log(\text{Bun})$ indicates Bun should enter linearly, not as its logarithm — so a linear term in Bun is appropriate.

2.3 Identifying poorly fitted observations

Type: Interpret figure

Two plots are used to look for outliers: an index plot of the deviance residuals, and a plot of the deviance residuals against the risk score.

(a) What does a large positive (or large negative) deviance residual indicate about an individual?

(b) Why are deviance residuals preferred over martingale residuals for these plots?

(c) What is the value of plotting the deviance residuals against the risk score rather than just making an index plot?

Solution.

(a) A large positive residual means the individual had an event sooner than expected under the model; a large negative residual means they survived longer than expected given their covariates. Either way they are poorly fitted (potential outliers).

(b) Deviance residuals are more symmetric, so plots are easier to interpret and extreme observations stand out more clearly.
(c) It relates poor fit to the covariates: for example, if the largest residuals occur at low risk scores, individuals predicted to be low-risk are dying unexpectedly early, hinting at a misspecified covariate effect.

3 Influential observations (Book §4.3)

3.1 Delta-betas and influence

Type: Concept & notation

Builds on: Chapter 3

(a) What is a delta-beta, $\Delta_i\hat{\beta}_j$, and what does a large absolute value indicate?
(b) On which residual are the delta-betas based?
(c) How does influence on a single parameter estimate differ from the influence summarised by the likelihood displacement LD_i or the diagnostic $|l_{\max}|$?

Solution.

(a) $\Delta_i\hat{\beta}_j \approx \hat{\beta}_j - \hat{\beta}_{j(i)}$ is the approximate change in the estimate of β_j when observation i is omitted. A large absolute value means that observation unduly influences that coefficient (and hence the corresponding hazard ratio).
(b) On the score residuals (in combination with the variance–covariance matrix of the estimates).
(c) Delta-betas assess influence on one specific coefficient. LD_i and $|l_{\max}|_i$ summarise influence on the whole set of parameter estimates (the risk score, and hence the fitted hazard). An observation can be uninfluential for each coefficient individually yet influential for the set.

4 Checking proportional hazards (Book §4.4)

4.1 The log-cumulative hazard plot

Type: Assumptions

Builds on: Chapters 1–3

For the Cox model, $\log H_i(t) = \beta' x_i + \log H_0(t)$.

(a) Explain why, under proportional hazards, the log-cumulative hazard curves for different groups should be parallel.
(b) Describe how the plot is constructed when the covariate of interest is continuous.
(c) What are the main limitations of this plot?

Solution.

(a) Under proportional hazards, $\log H_i(t)$ differs between individuals only by the constant $\beta' x_i$, so plots of $\log \hat{H}(t)$ against $\log t$ for different groups differ only by a constant vertical shift — they are parallel. Non-parallel curves indicate non-proportional hazards.

(b) First categorise the continuous variable (e.g. into quartile-based groups), compute the Kaplan–Meier estimate for each group, and plot $\log \hat{H}(t) = \log(-\log \hat{S}(t))$ against $\log t$ for each group.
(c) It can only handle one or a few factors with enough observations per level; the curves are based on very few observations at late times; and it cannot adjust for other covariates. Schoenfeld-residual methods are preferred in more structured settings.

4.2 Schoenfeld residuals and the zph test

Type: Interpret R output

A Cox model with three variables (`treat`, `age`, `sex`) is fitted, and the proportional hazards assumption is tested. The output is:

```
          chisq df     p
treat      9.84  1 0.002
age        1.32  1 0.25
sex        0.66  1 0.42
GLOBAL    11.82  3 0.008
```

(a) What hypothesis does each row test?
(b) Which variable(s) show evidence against proportional hazards?
(c) What is the "GLOBAL" test, and what do you conclude overall?

Solution.

(a) Each row tests that the coefficient of that variable is constant over time — i.e. that proportional hazards holds for that variable. Under the null the statistic has a chi-squared distribution on 1 degree of freedom.
(b) `treat`: $\chi^2 = 9.84$, $P = 0.002$ — strong evidence its effect is time-dependent. `age` ($P = 0.25$) and `sex` ($P = 0.42$) show no evidence against proportional hazards.
(c) The global test jointly tests proportional hazards for all variables ($\chi^2 = 11.82$ on 3 d.f., $P = 0.008$), rejecting the assumption overall — driven by the time-dependence of the treatment effect.

4.3 What to do when proportional hazards fails

Type: What if …?

Suppose the zph test shows a treatment effect that decreases over time — a strong early benefit that wanes.

(a) What does it mean, substantively, that the treatment effect is time-dependent?
(b) Describe one model-based way (from §4.4.4) to capture this, and state the key added term.
(c) What does a test of that added term's coefficient amount to?

Solution.

(a) The hazard ratio comparing treatments is not constant: the treatment's benefit (or harm) is larger at some times than others. A single hazard ratio would misrepresent the effect.
(b) Add a time-dependent variable $X_2 = X_1 t$ (treatment $\times$ time) to the model, so the log hazard ratio becomes $\beta_1 + \beta_2 t$; β_2 captures how the effect changes with time. Often $X_1(t - t_0)$ is used instead, so β_1 is the log hazard ratio at a reference time t_0.
(c) Testing $\beta_2 = 0$ is a formal test of proportional hazards: if $\beta_2 = 0$ the effect does not depend on time.

5 Putting it together

5.1 A misleading diagnostic conclusion

Type: What's wrong here?

A researcher reports: "I checked my Cox model by plotting the Cox–Snell residuals against survival time; they scattered roughly symmetrically about zero, so the model fits well."

Explain what is wrong with this statement.

Solution. Three things are wrong. First, Cox–Snell residuals are non-negative and skew, centred on 1 (they approximate a unit exponential distribution), so they are never symmetric about zero. Second, plotting them against survival time (or its rank order) is unhelpful because they are correlated with time. Third, the appropriate overall check is the cumulative hazard plot of the residuals (a straight line of unit slope), supplemented by deviance-residual plots, functional form checks with martingale residuals, and a test of proportional hazards.

6 Mini-lab: checking a Cox model in R

6.1 Detecting non-proportional hazards

Type: Mini-lab

Run the code below. It simulates 400 patients in two treatment groups. The control group has a constant hazard, but the treated group has a hazard that *increases* over time, so the treatment effect is not constant. A Cox model is fitted and the proportional hazards assumption is tested.

```
set.seed(731)
n <- 400
treat <- rep(c(0, 1), each = n/2)
t0 <- rexp(n/2, rate = 0.05)                    # control: constant hazard
t1 <- rweibull(n/2, shape = 2, scale = 1/0.07)  # treated: increasing hazard
t <- c(t0, t1)
cens <- runif(n, 0, 60)
obs <- pmin(t, cens)
status <- as.numeric(t <= cens)
d <- data.frame(time = obs, status = status, treat = treat)
```

```r
library(survival)
fit <- coxph(Surv(time, status) ~ treat, data = d)
summary(fit)
```

```
Call:
coxph(formula = Surv(time, status) ~ treat, data = d)

  n= 400, number of events= 288

        coef exp(coef) se(coef)     z Pr(>|z|)
treat 0.5322    1.7027   0.1281 4.156 3.24e-05 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

      exp(coef) exp(-coef) lower .95 upper .95
treat     1.703     0.5873     1.325     2.188

Concordance= 0.508  (se = 0.018 )
Likelihood ratio test= 17.56  on 1 df,   p=3e-05
Wald test            = 17.27  on 1 df,   p=3e-05
Score (logrank) test = 17.62  on 1 df,   p=3e-05
```

```r
cox.zph(fit)
```

```
       chisq df       p
treat   63.6  1 1.5e-15
GLOBAL  63.6  1 1.5e-15
```

(a) From `summary(fit)`, report the estimated hazard ratio for `treat` and its p-value. What would a naive interpretation conclude?
(b) Report the test statistic and p-value from `cox.zph(fit)`. What does it tell you?
(c) Explain why the hazard ratio from (a) is misleading, and say what you should do next.

Solution.

(a) The estimated hazard ratio is about 1.70 ($P \approx 3 \times 10^{-5}$), so the naive Cox model suggests treatment is associated with a significantly higher hazard.
(b) `cox.zph` gives $\chi^2 = 63.6$ on 1 degree of freedom, $P \approx 1.5 \times 10^{-15}$ — overwhelming evidence against proportional hazards. The treatment effect is strongly time-dependent.
(c) Because the hazards are not proportional (they actually cross), the single "average" hazard ratio is misleading: it averages a time-changing effect. The next step is to model the time-

dependence — for example, add a treatment × time term, or stratify — and describe how the effect varies over time rather than reporting one hazard ratio.